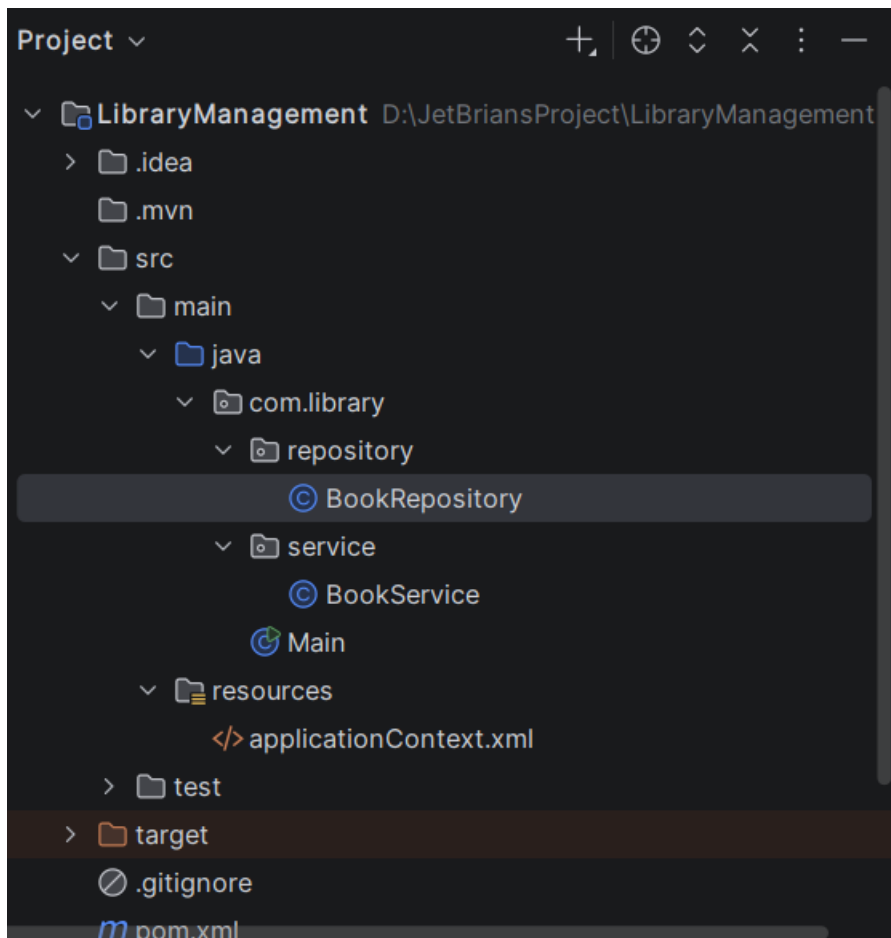


## Exercise 1: Configuring a Basic Spring Application

**Scenario:** Your company is developing a web application for managing a library. You need to use the Spring Framework to handle the backend operations.

### Folder Structure:



### BookRepository.java

```
package com.library.repository;

public class BookRepository {

    public void displayBooks() {
        System.out.println("Book Repository is working.");
    }

}
```

## BookService.java

```
package com.library.service;

import com.library.repository.BookRepository;

public class BookService {

    private BookRepository bookRepository;

    public void setBookRepository(BookRepository bookRepository) {
        this.bookRepository = bookRepository;
    }

    public void showService() {
        System.out.println("Book Service is working.");
        bookRepository.displayBooks();
    }

}
```

## Main.java

```
package com.library;

import com.library.service.BookService;
import org.springframework.context.ApplicationContext;
import org.springframework.context.support.ClassPathXmlApplicationContext;

public class Main {

    public static void main(String[] args) {

        ApplicationContext context =
            new ClassPathXmlApplicationContext("applicationContext.xml");

        BookService service =
            context.getBean("bookService", BookService.class);

        service.showService();

    }

}
```

## applicationContent.xml:

```
<?xml version="1.0" encoding="UTF-8"?>

<beans xmlns="http://www.springframework.org/schema/beans"
       xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"

       xsi:schemaLocation="
           http://www.springframework.org/schema/beans
```

```

    https://www.springframework.org/schema/beans/spring-beans.xsd">

    <bean id="bookRepository"
        class="com.library.repository.BookRepository"/>

    <bean id="bookService"
        class="com.library.service.BookService">

        <property name="bookRepository"
            ref="bookRepository"/>

    </bean>

</beans>

```

## Output:

```

package com.library;

import com.library.service.BookService;
import org.springframework.context.ApplicationContext;
import org.springframework.context.support.ClassPathXmlApplicationContext;

public class Main {

    public static void main(String[] args) {

        ApplicationContext context =
            new ClassPathXmlApplicationContext("applicationContext.xml");

        BookService service =
            context.getBean("bookService", BookService.class);

        service.showService();
    }
}

```

```

C:\Program Files\Eclipse Adoptium\jdk-17.0.18-hotspot\bin\java.exe ...
Book Service is working.
Book Repository is working.
Process finished with exit code 0

```